Security Log Parser Goal 3 Portfolio Report

Evidence for Foundational Skills in Software Development Tools

Field	Details
Student	Dongwoo Won
Student ID	25623683
Subject	UTS 43030 Professional Practice in Computing
Evidence goal	Goal 3 Demonstrate foundational skills in Software Development Tools
Project	Security Log Parser
Verification date	8 November 2026
Feature branch	feature/source-failure-summary

This report documents a working, reproducible Python command-line project, the completed source failure summary modification, and evidence produced by actual AI-run test, execution and Git commands. It also identifies what remains for the student to complete personally.

Purpose and evidence conclusion

Security Log Parser provides concrete evidence for most project-based parts of Goal 3 in the supplied Professional Learning Plan. The package contains a standard-library Python application, a command-line interface, automated tests, fictional sample data, JSON and CSV output, reproducible instructions, safe-data guidance and actual execution records. Modification exercise 2 adds failure totals for each login source while preserving the existing user totals.

Eighteen automated tests passed after the modification. The verified sample run processed eight valid events, counted two successful and six failed logins, produced four source failure totals, skipped two malformed lines, and detected one user whose three failures occurred within an inclusive 10-minute window.

Requirement status

Requirement basis	Status	Evidence or qualification
Goal 3 areas in the Learning Plan	Verified	Mapped to source, tests, CLI workflow, documentation and clean reproduction in this package.
Working security log parser	Verified	Source and fictional inputs are included and actual runs completed successfully.
Tests and README	Verified	Eighteen unittest cases passed; README covers the new console, JSON and source CSV output.
Current feature branch and commits	Verified	Git records a current baseline and feature commit. Codex executed them using the existing configured identity.
Git history authored through Dongwoo's own actions	Pending	No claim is made that Dongwoo personally ran the Git commands or authored the AI implementation.
Personal reflection	Pending	Prompts are included for Dongwoo to answer from direct experience.
Software Development Tools module requirements	Unverified	The Learning Plan refers to the module, but the module guide itself was not present in the supplied attachment.
Module completion and learning hours	Not claimed	This package contains no invented completion record or hours.

Source basis

The supplied Professional Learning Plan identifies Goal 3 as foundational skills in Software Development Tools. Page 2 lists Git-based version control and branching, Linux command-line workflow, debugging and unit testing, dependency and environment management, secure coding, README documentation and reproducible setup. It proposes a small Python security log-parser project and names working code, Git history, tests, a README and reflection as success evidence.

Project design and structure

The implementation separates data parsing, event analysis, export and command-line behavior. This makes each responsibility small enough to explain and test. The runtime uses only Python 3.9 or later and the standard library, so a package installation step is unnecessary.

Component	Responsibility
security_log_parser/parser.py	Reads UTF-8 CSV records, validates field count, timestamps and statuses, and records non-fatal warnings.
security_log_parser/analysis.py	Counts outcomes per user and source, then applies the inclusive user sliding-window failure rule.
security_log_parser/exporters.py	Writes detailed JSON plus separate per-user and per-source CSV reports.
security_log_parser/cli.py	Validates arguments, handles file errors, prints the summary and chooses output paths.
tests	Covers parsing, analysis, time boundaries, configuration, CLI errors, exports and source summary edge cases.
data	Provides fictional sample input and an empty input file.
evidence	Preserves raw command output, numbered captions and terminal-style captures.
README.md	Explains setup, commands, output, Git, debugging, limitations and safe handling.

Input and detection rules

- Each record contains a timezone-aware ISO 8601 timestamp, a fictional user, SUCCESS or FAILURE, and a fictional source.
- Blank lines, comments and the optional header are ignored. Invalid records are skipped and reported with their line number.
- Only FAILURE events enter detection. Events are grouped per user and sorted by time.
- The time boundary is inclusive. A span exactly equal to the configured window qualifies; a larger span does not.
- The tool reports the first qualifying window per user and exports user and source summaries to JSON and separate CSV files.

Baseline execution results

Baseline automated tests

Before the source-summary modification, the existing suite completed 14 tests with no failures. It established the baseline for normal and empty inputs, malformed records, missing files, exports, user counts, threshold behavior, exact and just-outside time boundaries, out-of-order events and invalid configuration. The expanded 18-test result is shown later with the feature evidence.



Figure 1 Baseline automated test run showing 14 passing tests

Debugging evidence

The first run failed because the initial data classes used the slots option, which is unavailable in Python 3.9. The fix removed that option and changed the documented minimum from Python 3.11 to Python 3.9. The original failed transcript remains in evidence/raw/debugging-first-run.txt. The successful test run followed the fix.

Sample analysis and error handling

The fictional sample contains eight valid events and two intentionally malformed lines. The program reported two successes and six failures. Sentinel-demo generated one alert because its three failures occurred at 09:10, 09:15 and 09:20, exactly filling the inclusive 10-minute boundary.



Figure 2 Sample log analysis with counts warnings and the inclusive boundary alert

Missing file behavior

A missing input file produces a concise message on standard error and exit status 2. This behavior can be detected by shell scripts and automated workflows.

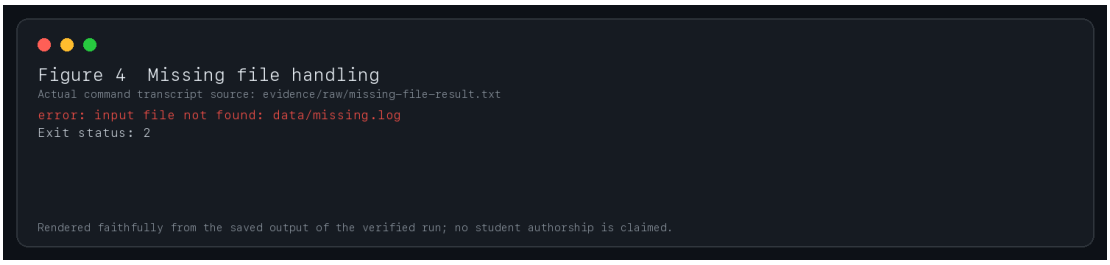


Figure 4 Missing file error and exit status 2

Source failure summary modification

Reason for the change

The original report grouped failed logins by user but did not show whether failures were concentrated on one login source. The modification adds a second view without changing the existing user report or the repeated-failure detection rule.

Implementation

- analyse_events counts only FAILURE events by source and sorts the output by source name.
- analysis.json now includes a sources array containing source and failed_logins fields.
- source_failure_summary.csv is written separately from user_summary.csv.
- The CLI prints Failures by source and lists all three output paths.
- Sources with only SUCCESS events are omitted. An input with no failures produces an empty JSON source list and a header-only source CSV.

Feature tests

Four new tests verify aggregation when two users share one source, exclusion of SUCCESS events, an empty source list when there are no failures, and creation of a header-only source CSV. The original 14 tests still pass, giving 18 passing tests in total.



Figure 6 Expanded feature test run showing 18 passing tests

Source summary execution evidence

The sample contains six failures across four fictional sources. The feature preserved all user totals and the user time-window alert, then reported source totals of 1, 2, 2 and 1. The separate CSV and JSON sources array contain the same values.

Figure 7 Example run with source counts

Actual command transcript: evidence/source-failure-summary/raw/example-run.txt

```
Security Log Parser
Input: data/sample_login.log
Valid events: 8
Successful logins: 2
Failed logins: 6
Failures by user:
  alex-demo: 1
  blair-demo: 2
  casey-demo: 0
  sentinel-demo: 3
Failures by source:
  training-laptop-01: 1
  training-laptop-02: 2
  training-laptop-04: 2
  training-laptop-05: 1
Repeated failures: threshold=3, window=10 minutes (inclusive boundary)
  ALERT sentinel-demo: 3 failures from 2026-08-01T09:10:00+10:00 to 2026-08-01T09:20:00+10:00
Malformed lines skipped: 2
warning: line 10: expected 4 fields, found 1
warning: line 11: status must be SUCCESS or FAILURE
JSON: results/analysis.json
User CSV: results/user_summary.csv
Source CSV: results/source_failure_summary.csv
```

Rendered from the saved output of the AI-run command; no student execution is claimed.



Figure 7 Example run with the added source failure totals

Figure 8 Source CSV and JSON output

Actual command transcript: evidence/source-failure-summary/raw/source-output-preview.txt

```
$ cat results/source_failure_summary.csv
source,failed_logins
training-laptop-01,1
training-laptop-02,2
training-laptop-04,2
training-laptop-05,1

$ selected JSON source data
{
  "summary": {
    "valid_events": 8,
    "successful_logins": 2,
    "failed_logins": 6,
    "users_with_repeated_failures": 1
  },
  "sources": [
    {
      "source": "training-laptop-01",
      "failed_logins": 1
    },
    {
      "source": "training-laptop-02",
      "failed_logins": 2
    },
    {
      "source": "training-laptop-04",
      "failed_logins": 2
    },
    {
      "source": "training-laptop-05",
      "failed_logins": 1
    }
  ]
}
```

Rendered from the saved output of the AI-run command; no student execution is claimed.



Figure 8 Separate source CSV and matching JSON source data

Git branch and committed change

The repository was initialised with a current baseline commit, then the modification was developed on feature/source-failure-summary. Commit 26b5588 records the implementation, tests, README changes and execution evidence. Git already contained an author name and email, so Codex used that configuration without changing or inventing identity values. Codex, not Dongwoo, executed these Git commands.



Figure 9 Current feature branch commits and implementation diff summary

Stored evidence

The evidence/source-failure-summary directory contains the baseline and final test transcripts, example run, no-failure run, JSON and CSV preview, pre-commit diff, committed patch, Git summary, captions and numbered PNGs.

Baseline exported evidence

Before the modification, the CSV contained one row per user with success and failure counts, the detection flag and the qualifying window. The JSON retained the configuration, totals, alerts and rejected-line details. The later feature evidence shows the added source CSV and matching JSON sources array.



Figure 3 Baseline user CSV and selected JSON fields

Clean environment reproduction

A clean project copy was created outside the working project without carrying over the generated results directory. All 18 tests passed, and the sample command reproduced the same user totals, four source totals and alert. It regenerated analysis.json, user_summary.csv and source_failure_summary.csv.

Figure 5 Clean environment reproduction

Actual command transcript source: evidence/raw/clean-environment-results.txt

```
$ python3 -m unittest discover -s tests -v
test_below_threshold_is_not_detected (test_analysis.AnalysisTests) ... ok
test_counts_success_and_failure_by_user (test_analysis.AnalysisTests) ... ok
test_exact_time_boundary_is_inclusive (test_analysis.AnalysisTests) ... ok
test_invalid_configuration_is_rejected (test_analysis.AnalysisTests) ... ok
test_just_outside_time_boundary_is_excluded (test_analysis.AnalysisTests) ... ok
test_out_of_order_events_are_sorted (test_analysis.AnalysisTests) ... ok
test_source_failures_combine_multiple_users (test_analysis.AnalysisTests) ... ok
test_source_summary_excludes_successes (test_analysis.AnalysisTests) ... ok
test_source_summary_is_empty_when_there_are_no_failures (test_analysis.AnalysisTests) ... ok
test_threshold_is_detected (test_analysis.AnalysisTests) ... ok
test_exports_json_and_csv (test_cli.CliTests) ... ok
test_missing_input_returns_error_code (test_cli.CliTests) ... ok
test_no_failures_creates_source_csv_with_header_only (test_cli.CliTests) ... ok
test_empty_input (test_parser.ParserTests) ... ok
test_malformed_lines_are_skipped_and_reported (test_parser.ParserTests) ... ok
test_missing_file_raises_file_not_found (test_parser.ParserTests) ... ok
test_naive_timestamp_is_rejected (test_parser.ParserTests) ... ok
test_normal_input (test_parser.ParserTests) ... ok

----------------------------------------------------------------------
Ran 18 tests in 0.005s

OK

$ python3 -m security_log_parser data/sample_login.log --output-dir results --threshold 3 --window-
minutes 10
warning: line 10: expected 4 fields, found 1
warning: line 11: status must be SUCCESS or FAILURE
Security Log Parser
Input: data/sample_login.log
Valid events: 8
Successful logins: 2
Failed logins: 6
Failures by user:
  alex-demo: 1
  blair-demo: 2
  casey-demo: 0
  sentinel-demo: 3
Failures by source:
  training-laptop-01: 1
  training-laptop-02: 2
  training-laptop-04: 2
  training-laptop-05: 1
Repeated failures: threshold=3, window=10 minutes (inclusive boundary)
  ALERT sentinel-demo: 3 failures from 2026-08-01T09:10:00+10:00 to 2026-08-01T09:20:00+10:00
Malformed lines skipped: 2
JSON: results/analysis.json
User CSV: results/user_summary.csv
Source CSV: results/source_failure_summary.csv

$ find results -maxdepth 1 -type f -print | sort
results/analysis.json
results/source_failure_summary.csv
results/user_summary.csv
```

Rendered faithfully from the saved output of the verified run; no student authorship is claimed.

Figure 5 Clean copy feature test run, sample analysis and three regenerated output files

Connection to Goal 3

Learning Plan area	Evidence in this package	Status for portfolio use
Git version control and branching	Current baseline and feature commits exist on feature/source-failure-summary; diff evidence is stored.	AI-executed workflow verified. Student-executed evidence remains to be created.
Linux command-line workflow	Run, test, inspect, compare and exit-status commands are documented. Raw transcripts record actual AI-run commands.	Technical evidence present; student should repeat and explain the commands.
Debugging and unit testing	Initial compatibility work is retained and the feature suite now contains 18 passing unittest cases.	Verified.
Dependency and environment management	No third-party runtime dependencies; Python version documented; clean-copy verification completed.	Verified for this project.
Secure coding	Strict schema and timestamp validation, non-execution of input, clear error handling and safe-data guidance.	Verified within the stated limitations.
README documentation	README documents the source console section, JSON array, separate CSV, no-failure behavior and limitations.	Verified.
Reproducible setup	A separate clean copy passed all tests and produced the expected reports.	Verified.
Software Development Tools module	Only the Learning Plan reference was available; no module instructions or completion record were supplied.	Compliance and completion unverified.

Evidence interpretation

The package demonstrates that the technical artifacts and AI-executed Git workflow work. It does not establish that Dongwoo personally performed the implementation, verification or commits, or completed the planned module. Strong final evidence should add a student-run command record, a further student-authored change, answers to the comprehension questions, and reflection based on direct work.

Limitations and safe data handling

Technical limitations

- The parser reads the file into memory and is not designed for continuous or very large streams.
- Only one four-field CSV schema and two status values are accepted.
- Detection is per user and reports the first qualifying window, not every overlapping window.
- The source report is a total count only. It does not apply a time window, count unique users or generate source-based alerts.
- Timestamps are validated syntactically but their trustworthiness is not verified.
- The alert is an educational indicator, not proof of malicious activity or a replacement for a security monitoring platform.
- Reports are unencrypted and files with the same output names are replaced.

Safe data handling

All supplied logs use fictional labels. Real authentication logs can contain names, device identifiers, IP addresses, locations and security-sensitive behavior. Before using real data, obtain authority, minimise fields, restrict access, follow retention rules, remove secrets and avoid committing logs or reports to Git. Rejected-line content also requires review before sharing because malformed records may still contain sensitive data.

Code safety decisions

- The program never evaluates or executes log content.
- Argument values are validated before analysis.
- Malformed records are isolated and reported instead of causing a partial crash.
- Timezone-aware timestamps reduce ambiguous boundary comparisons.
- There are no network calls or external runtime dependencies.

AI contribution and student responsibility

OpenAI Codex created the initial source code, tests, fictional sample data, README, evidence captures and report structure. Codex also implemented modification exercise 2, created the feature branch, ran the tests and examples, and executed the current commits using the pre-existing Git author configuration.

Dongwoo Won should review and understand every submitted artifact, run the project independently, complete a further personal modification or reimplement part of this feature, create Git evidence through personal actions, answer the comprehension questions, and write reflection responses from direct experience. The final submission should not imply that Dongwoo authored the AI-generated implementation or personally performed AI-run commands.

Claims excluded from this report

- No past Git dates were created and the current commits are not described as student-executed work.
- No learning hours were recorded as completed.
- No Software Development Tools module completion was claimed.
- No personal reflection was invented.
- No real login or personal data was used.

Reflection prompts for Dongwoo

1. Which command or file was hardest to understand when you ran the project, and what did you do to resolve that difficulty?
2. What failure did you encounter while completing your own modification, and how did a test or debugger help you locate it?
3. How did using a Git branch and reviewing the diff change the way you managed the modification?
4. What did the exact 10-minute boundary test teach you about translating a written rule into code?
5. Which secure-data practice would matter most if the fictional input were replaced with a real authentication log, and why?
6. What would you change in your next software project to make its setup and evidence more reproducible?

Response note: answer these prompts only after completing and recording the work yourself.

Student exercises and comprehension check

Code modification exercises

1. Add a --status-case-sensitive option. Test both default and strict behavior, then explain the compatibility trade-off.
2. Completed by Codex: add a per-source failure summary to JSON and CSV on a feature branch with shared-source tests. Dongwoo should review, rerun and explain it rather than claim initial authorship.
3. Add a --redact-usernames option using stable SHA-256-based aliases. Explain why unsalted hashing may still allow guessing and what a protected salt changes.

Comprehension questions

1. Why does the parser require a UTC offset in every timestamp?
2. What condition in the sliding-window logic makes the time boundary inclusive?
3. Why are malformed lines reported instead of silently ignored?
4. How does the analysis exclude successful logins from repeated-failure detection?
5. What is the difference between the threshold and the time window?
6. Why can a passing test suite still be insufficient evidence that the program is correct?
7. What does exit status 2 communicate to a shell script?
8. Which changes should be reviewed before a Git commit, and why?
9. What sensitive information could appear in real authentication logs even when passwords are absent?
10. Which limitation would you address first for a large production log, and what design would you consider?

Submission checklist

- Run the tests and sample command personally and retain the output.
- Complete one exercise with current, accurate Git commits.
- Answer the comprehension and reflection questions in your own words.
- Check that the final package still contains only fictional data.
- Confirm submission rules against the actual Software Development Tools module guide when it is available.